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Topic 9 — Separate Chemistry 2

Edexcel GCSE (9-1) Chemistry (1CH0) — Paper 2 — chemistry-only content

This topic brings together four areas that appear only in GCSE Chemistry: **qualitative analysis** (identifying ions with chemical tests and instruments), **organic chemistry** beyond crude oil (alkanes, alkenes, alcohols and carboxylic acids), **polymers** (addition, condensation and natural polymers), and the **bulk and surface properties of materials**, including nanoparticles.

The whole of Topic 9 is chemistry-only — it is not in Combined Science (in the specification every point is labelled with a "C"). It contains **two core practicals**: identifying the ions in unknown salts, and comparing the temperature rise produced by burning different alcohols.

9.1 Tests for Cations — Flame Tests

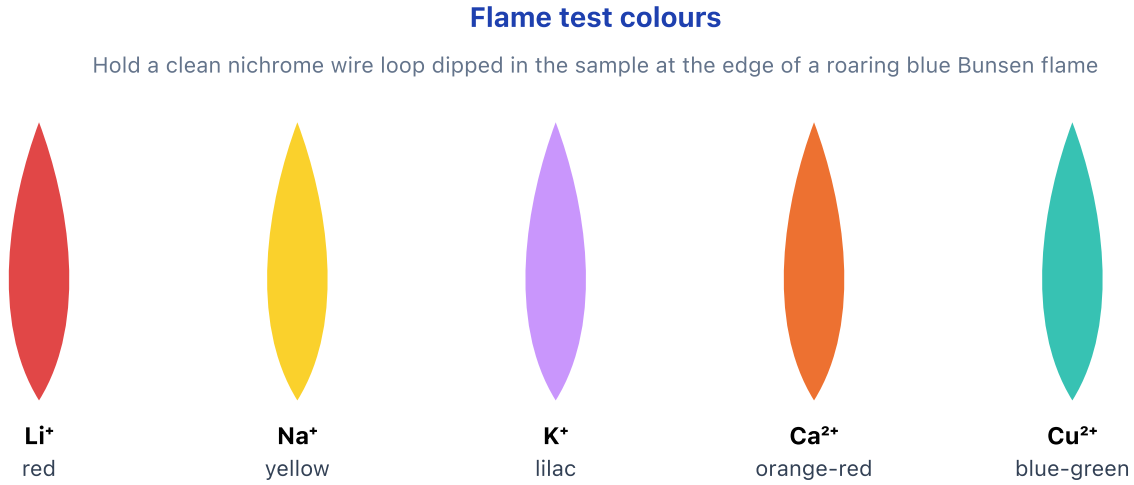
> The test for any ion must be **unique**: it must give a result that **no other ion** gives, otherwise a positive result could not tell you which ion is present. For example, calcium and aluminium ions both give a white precipitate with a few drops of sodium hydroxide, so you must go on to add excess (or use a flame test) to tell them apart; adding dilute acid first in the anion tests removes carbonate ions that would otherwise also give a precipitate.

Many metal ions produce a characteristic flame colour when their compounds are heated strongly. The heat excites electrons, which then release energy as light of particular colours.

Metal ion	Flame colour
Lithium, Li^+	**red** (crimson)
Sodium, Na^+	**yellow** (orange-yellow)

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Potassium, K^+	**lilac**
Calcium, Ca^{2+}	**orange-red** (brick red)
Copper, Cu^{2+}	**blue-green**



Method

1. Clean a **nichrome** (or platinum) wire loop by dipping it in **concentrated hydrochloric acid** and holding it in a **blue** Bunsen flame until it gives no colour.
2. Dip the loop in the acid again, then into the solid sample so some sticks to it.
3. Hold the loop at the **edge of a blue (roaring) flame** and observe the colour.
4. Clean the wire thoroughly between tests so samples don't contaminate each other.

> **Exam Tip:** Limitations of flame tests: if a sample contains a **mixture** of metal ions, one colour can **mask** another (sodium's intense yellow hides most others). Some ions give no flame colour, and colours are judged by eye (subjective). This is one reason instrumental methods such as flame photometry are useful (9.4).

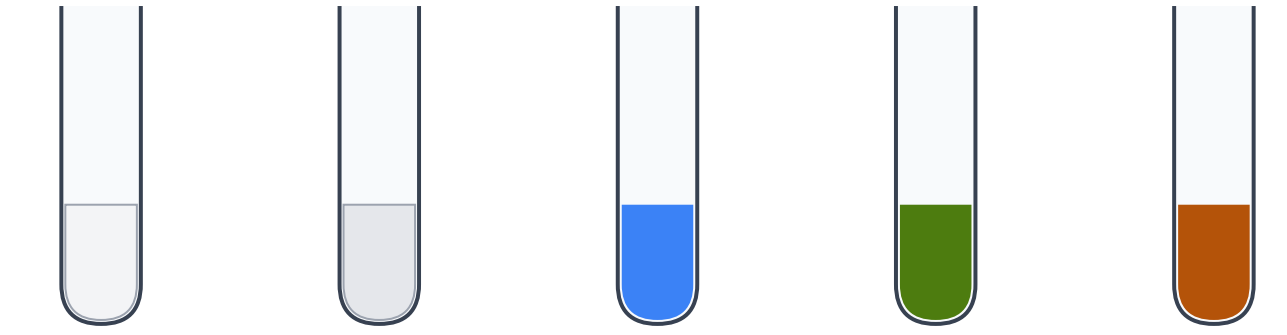
9.2 Tests for Cations — Sodium Hydroxide Solution

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Add a few drops of **dilute sodium hydroxide solution** to a solution of the unknown compound. Many metal ions form an insoluble **metal hydroxide precipitate** with a characteristic colour.

Ion	Observation with a few drops of NaOH(aq)	With excess NaOH(aq)	Ionic equation **(HT)**
Aluminium, Al^{3+}	**white** precipitate	precipitate **dissolves** → colourless solution	$Al^{3+}(aq) + 3OH^-(aq) \rightarrow Al(OH)_3(s)$
Calcium, Ca^{2+}	**white** precipitate	precipitate **remains**	$Ca^{2+}(aq) + 2OH^-(aq) \rightarrow Ca(OH)_2(s)$
Copper(II), Cu^{2+}	**blue** precipitate	remains	$Cu^{2+}(aq) + 2OH^-(aq) \rightarrow Cu(OH)_2(s)$
Iron(II), Fe^{2+}	**green** precipitate (turns brown on standing in air as it oxidises)	remains	$Fe^{2+}(aq) + 2OH^-(aq) \rightarrow Fe(OH)_2(s)$
Iron(III), Fe^{3+}	**orange-brown** (rust-coloured) precipitate	remains	$Fe^{3+}(aq) + 3OH^-(aq) \rightarrow Fe(OH)_3(s)$

Metal hydroxide precipitates formed with NaOH(aq)



Al^{3+}
white;
dissolves in excess

Ca^{2+}
white;
stays in excess

Cu^{2+}
blue

Fe^{2+}
green

Fe^{3+}
orange-brown

Test for the ammonium ion, NH_4^+

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1. Add sodium hydroxide solution to the sample and **warm gently**.
2. Hold **damp red litmus paper** at the mouth of the test tube.
3. If NH_4^+ is present, **ammonia gas** is given off, which turns damp red litmus paper **blue** (ammonia is alkaline). It also has a sharp, pungent smell.

$$NH_4^+(aq) + OH^-(aq) \rightarrow NH_3(g) + H_2O(l)$$

Exam Tip: Al^{3+} and Ca^{2+} both give a white precipitate with a few drops of NaOH — to tell them apart, add **excess** NaOH ($Al(OH)_3$ dissolves; $Ca(OH)_2$ does not) or do a **flame test** (Ca^{2+} gives orange-red; Al^{3+} gives no characteristic colour). The litmus paper for ammonia must be **damp** — gases need to dissolve in water to affect an indicator.

9.3 Tests for Anions

Anion	Test	Positive result	Ionic equation **(HT)**
Chloride, Cl^-	Add **dilute nitric acid**, then **silver nitrate solution**	**white** precipitate (AgCl)	$Ag^+(aq) + Cl^-(aq) \rightarrow AgCl(s)$
Bromide, Br^-		**cream** precipitate (AgBr)	$Ag^+(aq) + Br^-(aq) \rightarrow AgBr(s)$
Iodide, I^-		**yellow** precipitate (AgI)	$Ag^+(aq) + I^-(aq) \rightarrow AgI(s)$
Sulfate, SO_4^{2-}	Add **dilute hydrochloric acid**, then **barium chloride solution**	**white** precipitate ($BaSO_4$)	$Ba^{2+}(aq) + SO_4^{2-}(aq) \rightarrow BaSO_4(s)$
Carbonate, CO_3^{2-}	Add **dilute acid**; bubble any gas through **limewater**	**effervescence**; gas turns limewater **milky (cloudy)** — CO_2	$CO_3^{2-}(s/aq) + 2H^+(aq) \rightarrow CO_2(g) + H_2O(l)$

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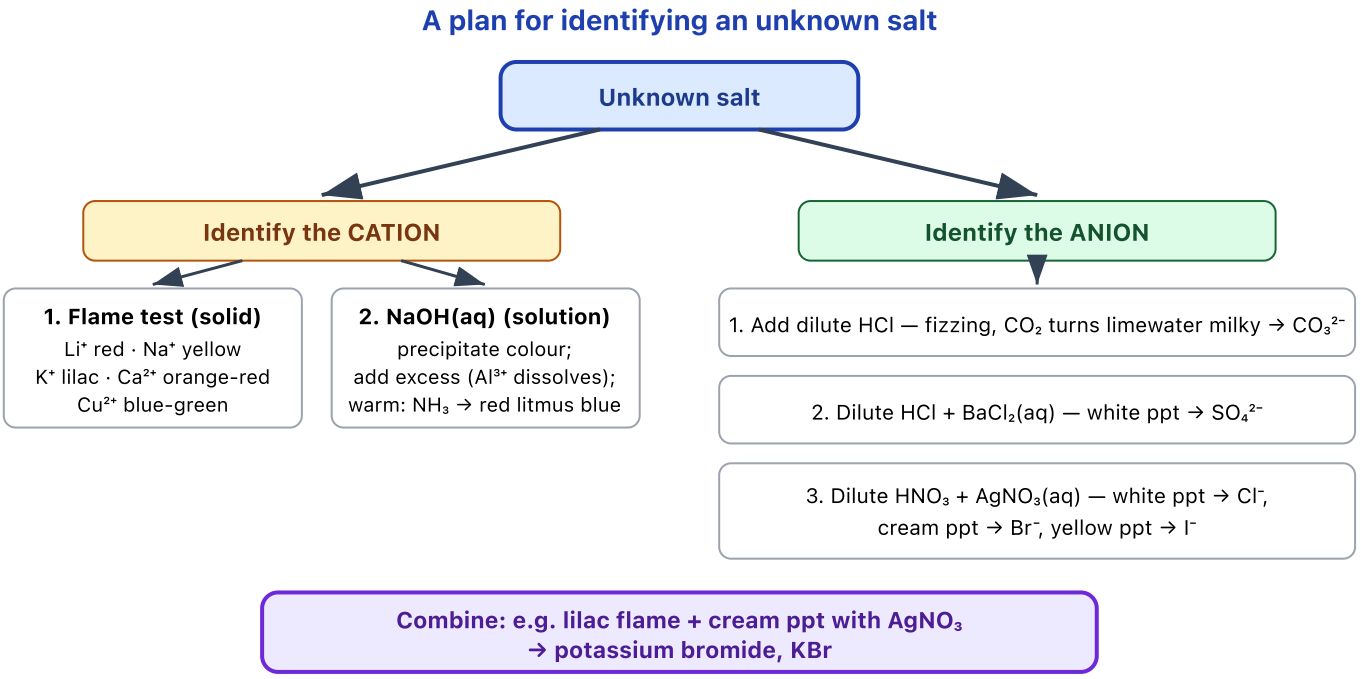
Why add acid first?

- In the halide and sulfate tests, the acid is added first to **remove carbonate ions**, which would otherwise also form a precipitate with silver or barium ions and give a false positive.
- For the halide test you must use **nitric acid**, not hydrochloric acid — HCl contains chloride ions, which would give a white precipitate whatever the sample.
- For the sulfate test you use **hydrochloric acid**, not sulfuric acid — H_2SO_4 contains sulfate ions.

> **Exam Tip:** A really common exam question: "Why is dilute nitric acid added before silver nitrate?" Answer: to react with/remove any carbonate ions, which would also give a precipitate. "Why not hydrochloric acid?" Because it adds chloride ions, giving a white precipitate even if the sample has no chloride.

9.4 Core Practical — Identifying the Ions in Unknown Salts

The core practical uses the tests above to identify the cation and anion in unknown salts.



Safety and good practice

- Wear eye protection; sodium hydroxide is corrosive and silver nitrate stains skin. Barium chloride is toxic. Concentrated HCl used to clean wires is corrosive.
- Use **small amounts** (a few drops) of reagents and add NaOH dropwise so you see the first precipitate before testing with excess.
- Use a fresh, clean test tube for every test to avoid contamination.

Worked Example: An unknown white solid gives an orange-red flame. A solution of it gives a white precipitate with a few drops of NaOH that does not dissolve in excess. With dilute HCl and $BaCl_2$ solution it gives a white precipitate. Identify the salt.
Orange-red flame + white precipitate insoluble in excess → **Ca^{2+}**. White precipitate with $BaCl_2$ → **SO_4^{2-}**. The salt is **calcium sulfate, $CaSO_4$**.

Worked Example: A green solution gives a green precipitate with NaOH(aq) that slowly turns brown at the surface. With dilute nitric acid and silver nitrate it gives a white precipitate. Identify the salt and **(HT)** write the ionic equation for the precipitate with NaOH.
Green precipitate → **Fe^{2+}**; white ppt with $AgNO_3$ → **Cl^-**. Salt = **iron(II) chloride, $FeCl_2$**.
$Fe^{2+}(aq) + 2OH^-(aq) \rightarrow Fe(OH)_2(s)$

9.5 Instrumental Methods and Flame Photometry

Instrumental methods of analysis use machines to detect and measure substances. Compared with simple chemical tests, they can improve:

- **sensitivity** — they can detect very small amounts of a substance,
- **accuracy** — results are reliable and not dependent on judging colours by eye,
- **speed** — tests are quick and can be automated, running many samples.

Disadvantages: the equipment is expensive, needs trained operators, and results usually have to be compared with reference data.

Flame photometry

In a **flame photometer**, a solution is sprayed into a flame. The metal ions emit light; the instrument measures the **wavelengths** of light given out and the **intensity** of that light. Each metal ion produces a characteristic **emission spectrum** (a pattern of lines of specific wavelengths).

You must be able to evaluate data from a flame photometer to:

1. **identify metal ions** by comparing the data (e.g. emission line wavelengths) with **reference data**. This works even in **mixtures**, because each ion's lines appear separately — unlike a flame test where colours mask each other;
2. **determine the concentration of ions** in dilute solution using a **calibration curve**.

Identifying ions by comparing emission spectra (illustrative)



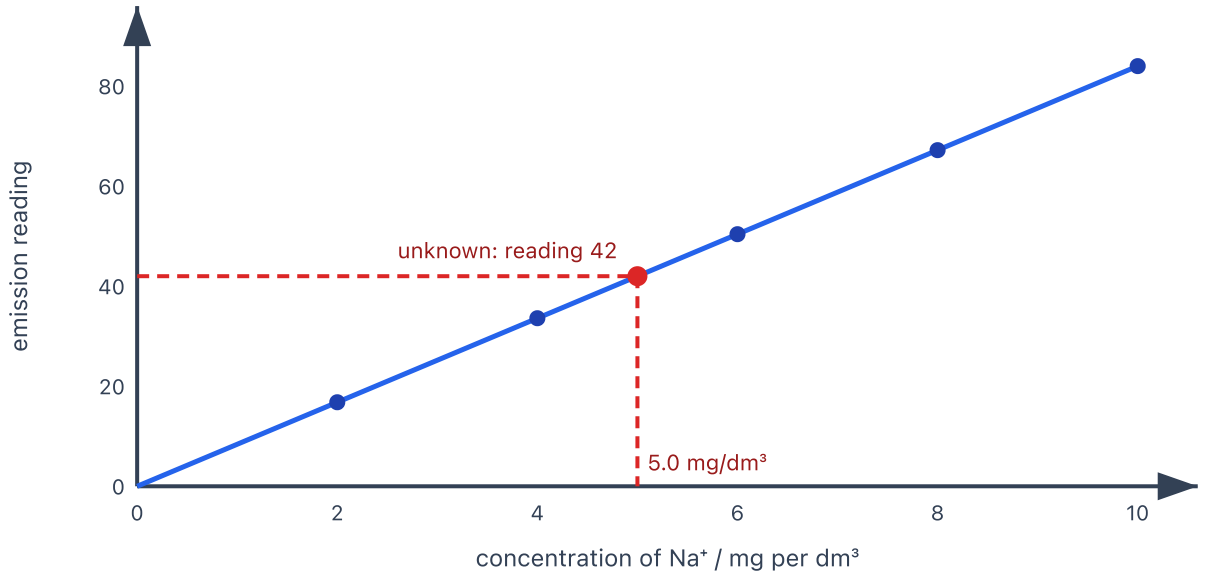
Every line in the sample matches a line of A or C → the sample contains ions A and C, not B.
wavelength →

Using a calibration curve

1. Make up a series of **standard solutions** of the metal ion with known concentrations.
2. Measure the emission intensity (reading) for each standard with the flame photometer.
3. Plot **reading (y)** against **concentration (x)** and draw a line of best fit — the calibration curve.
4. Measure the reading for the unknown solution, then read across to the line and down to find its concentration.

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Calibration curve for Na^+ (illustrative data)



Worked Example: *(Illustrative data.)* Using the calibration curve above, a sample of mineral water gives a reading of 42. Read across from 42 to the line, then down: concentration of Na^+ = **5.0 mg per dm³**. If the sample had been diluted 10 times before measuring, the original concentration would be 5.0 × 10 = **50 mg per dm³**.

Exam Tip: A calibration curve only gives reliable values **within the range** of the standards measured — don't extrapolate far beyond it. If the reading is too high, dilute the sample by a known factor and multiply back.

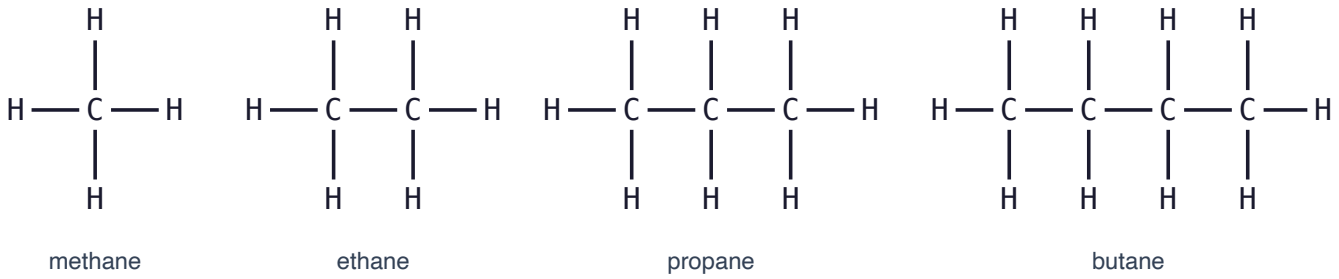
9.6 Alkanes and Alkenes

Alkanes — saturated hydrocarbons

Alkanes are hydrocarbons containing only **single C–C bonds**. They are **saturated**: each carbon atom forms four single bonds, so no more atoms can be added to the molecule. General formula $\mathbf{C_nH_{2n+2}}$.

Name	Molecular formula	Structural formula
methane	CH_4	CH_4
ethane	C_2H_6	CH_3CH_3
propane	C_3H_8	$CH_3CH_2CH_3$
butane	C_4H_{10}	$CH_3CH_2CH_2CH_3$

Displayed formulae of the first four alkanes



Alkenes — unsaturated hydrocarbons

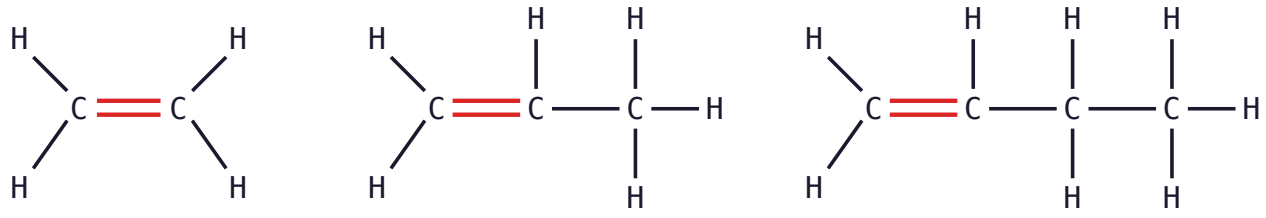
Alkenes are hydrocarbons containing a **carbon–carbon double bond, C=C**. They are **unsaturated**: the double bond can open up so that more atoms can be added. The **C=C double bond is the functional group** of the alkenes. General formula $\mathbf{C_nH_{2n}}$.

Name	Molecular formula	Structural formula
ethene	C_2H_4	$CH_2=CH_2$
propene	C_3H_6	$CH_2=CHCH_3$
butene (but-1-ene)	C_4H_8	$CH_2=CHCH_2CH_3$

There is no "methene" — an alkene needs at least two carbon atoms to have a C=C bond. Butene has isomers (the double bond can be in different positions); you only need to draw one correct structure.

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Displayed formulae of alkenes (C=C in red)



Addition reaction with bromine — the test for alkenes

Alkenes undergo **addition reactions**: the C=C double bond opens and atoms add across it, forming a saturated product. With bromine:

$$C_2H_4 + Br_2 \rightarrow C_2H_4Br_2 \quad \text{(ethene + bromine → 1,2-dibromoethane)}$$

$$CH_2{=}CH_2 + Br{-}Br \rightarrow CH_2Br{-}CH_2Br$$

Test for an alkene (C=C): shake the compound with **bromine water** (orange). An alkene **decolourises** it — orange → **colourless**. An alkane does not react, so the bromine water **stays orange**.

Bromine water test



alkane + bromine water
stays orange (no reaction)

alkene + bromine water
orange → colourless

Exam Tip: Say "**colourless**", never "clear". Clear means see-through (orange bromine water is clear too!). Bromine water decolourising shows a C=C bond is present — the molecule is unsaturated.

Combustion of alkanes and alkenes

The **complete combustion** of alkanes and alkenes involves the **oxidation** of the hydrocarbons to produce **carbon dioxide and water**:

$$C_3H_8 + 5O_2 \rightarrow 3CO_2 + 4H_2O \quad \text{(propane)}$$

$$C_2H_4 + 3O_2 \rightarrow 2CO_2 + 2H_2O \quad \text{(ethene)}$$

Alkenes tend to burn with a **smokier flame** than alkanes in air, because they have a higher proportion of carbon, so incomplete combustion is more likely. (Incomplete combustion is covered in Topic 8.)

	Alkanes	Alkenes
General formula	C_nH_{2n+2}	C_nH_{2n}
Bonding	C–C single bonds only	contain one C=C double bond
Saturated?	saturated	unsaturated
Functional group	none (only C–C, C–H)	C=C
Bromine water	stays orange	decolourised
Reactivity	fairly unreactive (combustion only at GCSE)	more reactive — addition reactions, polymerisation
Main use	fuels	making polymers and other chemicals

9.7 Alcohols

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Alcohols are a homologous series of compounds containing the **–OH functional group** (hydroxyl group). The general formula is $C_nH_{2n+1}OH$.

Alcohol	Molecular formula	Structural formula
methanol	CH_3OH	CH_3OH
ethanol	C_2H_5OH	CH_3CH_2OH
propanol (propan-1-ol)	C_3H_7OH	$CH_3CH_2CH_2OH$
butanol (butan-1-ol)	C_4H_9OH	$CH_3CH_2CH_2CH_2OH$

Only the straight-chain alcohols with –OH on the end carbon (propan-1-ol, butan-1-ol) are required. Displayed formulae: draw the alkane chain and replace one end H with –O–H, e.g. ethanol: H–C(H)(H)–C(H)(H)–O–H.

Alcohols can be **dehydrated** (water removed) to form **alkenes**, e.g. ethanol → ethene + water: $C_2H_5OH \rightarrow C_2H_4 + H_2O$.

Exam Tip: When drawing the displayed formula of an alcohol, show the **O–H bond** (draw –O–H, not just –OH, if the question says "displayed"). The –OH in an alcohol is **covalently bonded** — alcohols are not alkalis and do not contain hydroxide ions.

Making ethanol by fermentation

A solution of ethanol is made by the **fermentation** of carbohydrates (sugars) using **yeast**:

$$C_6H_{12}O_6(aq) \rightarrow 2C_2H_5OH(aq) + 2CO_2(g)$$

1. Dissolve a sugar (e.g. glucose or sucrose) in water and add **yeast**.
2. Keep the mixture warm — about **30–40 °C** (the optimum temperature for the yeast enzymes; above this the enzymes are denatured, below it the reaction is too slow).
3. Exclude air (**anaerobic** conditions) — e.g. use an air lock. With oxygen present, the yeast respires aerobically and microbes can oxidise the ethanol to ethanoic acid.

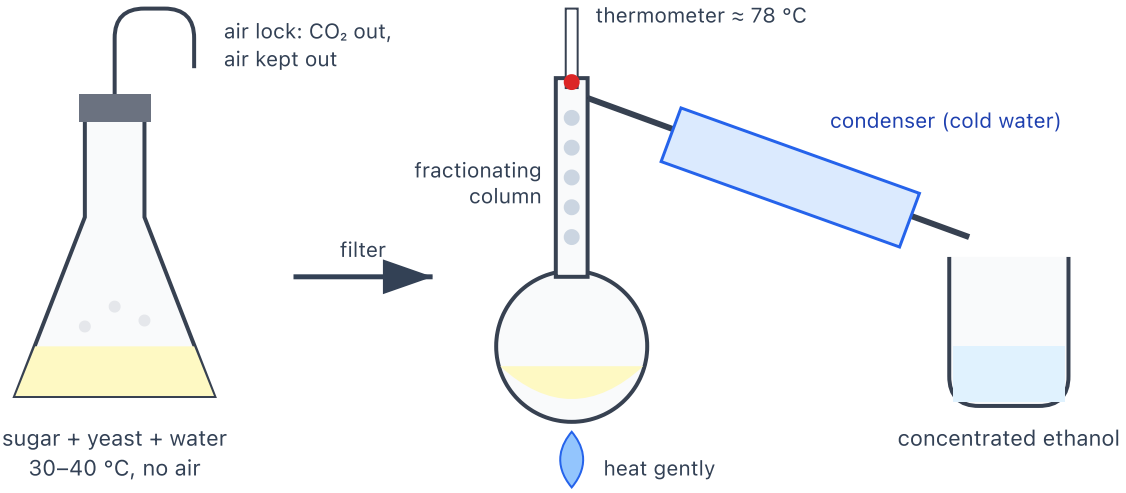
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4. The reaction is complete when bubbles of CO_2 stop. Fermentation stops when the ethanol concentration reaches about 15% because the ethanol kills the yeast.
5. Remove the yeast by **filtration**.

Concentrating ethanol by fractional distillation

The dilute ethanol solution is **concentrated by fractional distillation**. Ethanol (b.p. 78 °C) has a lower boiling point than water (100 °C), so it evaporates first, rises up the fractionating column and is condensed and collected. The temperature at the top of the column stays at about 78 °C while ethanol is distilling. This is how spirits (e.g. whisky, vodka) are made, and how ethanol is concentrated for use as a fuel.

From sugar to concentrated ethanol



Oxidation of ethanol to ethanoic acid

Ethanol can be **oxidised** to **ethanoic acid**, either by **microbes** (bacteria) in air, or by chemical oxidising agents (e.g. acidified potassium dichromate, which turns from orange to green). This is why an opened bottle of wine left exposed to air turns sour — **vinegar** is a dilute solution of ethanoic acid.

$$CH_3CH_2OH + 2[O] \rightarrow CH_3COOH + H_2O \quad ([O] = \text{oxygen from the oxidising agent})$$

9.8 Carboxylic Acids

Carboxylic acids are a homologous series containing the **–COOH functional group** (carboxyl group). General formula $C_nH_{2n+1}COOH$.

Carboxylic acid	Formula	Carbon atoms	Note
methanoic acid	HCOOH	1	in ant and nettle stings
ethanoic acid	CH_3COOH	2	vinegar
propanoic acid	C_2H_5COOH (CH_3CH_2COOH)	3	food preservative
butanoic acid	C_3H_7COOH ($CH_3CH_2CH_2COOH$)	4	smell of rancid butter

Exam Tip: Count the carbons carefully — the carbon in –COOH is included in the name. Butanoic acid has **four** carbons: C_3H_7COOH, not C_4H_9COOH. The displayed –COOH group is C double-bonded to one O and single-bonded to an O–H.

Reactions of carboxylic acids

Carboxylic acids dissolve in water to form **acidic solutions** (pH below 7). They are **weak acids** — only partially ionised in water — so they have a higher pH than a strong acid of the same concentration and react more slowly. They show typical acid reactions:

- With **metal carbonates** → salt + water + carbon dioxide (fizzing):

  $$2CH_3COOH + Na_2CO_3 \rightarrow 2CH_3COONa + H_2O + CO_2$$

  (the salt is sodium ethanoate)
- With **reactive metals** → salt + hydrogen: $2CH_3COOH + Mg \rightarrow (CH_3COO)_2Mg + H_2$
- With **alkalis** → salt + water: $CH_3COOH + NaOH \rightarrow CH_3COONa + H_2O$

Homologous series	Functional group	Ending of name	Example
alkanes	none	-ane	ethane, C_2H_6

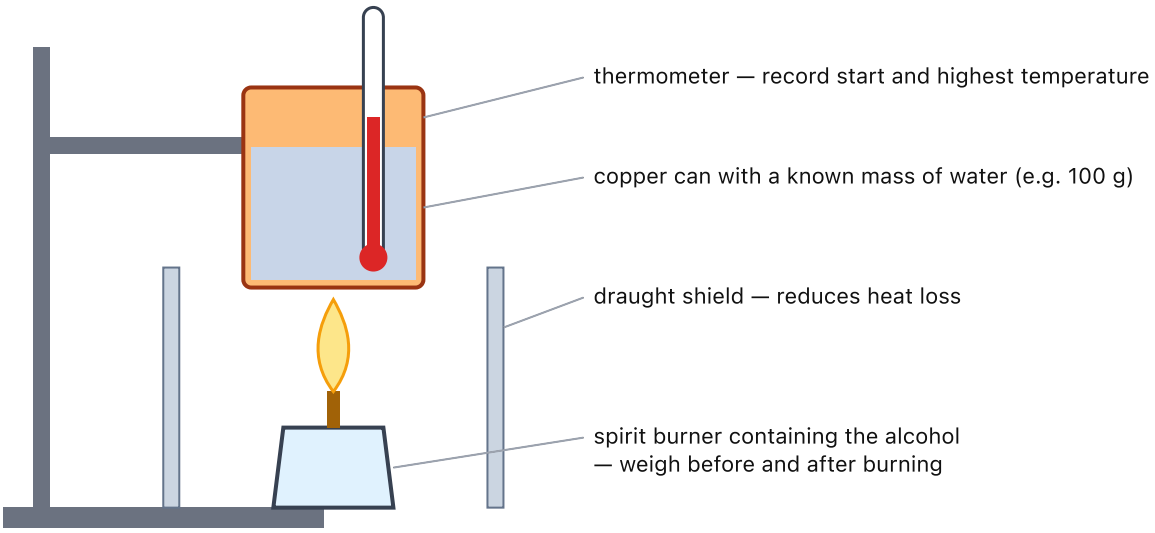
alkenes	C=C	-ene	ethene, C_2H_4
alcohols	–OH	-ol	ethanol, C_2H_5OH
carboxylic acids	–COOH	-oic acid	ethanoic acid, CH_3COOH

Members of the same homologous series have **similar chemical reactions** because their molecules contain the **same functional group**. So you can predict products for other members: e.g. propanol can be oxidised to propanoic acid, and propanoic acid reacts with sodium carbonate to give sodium propanoate, water and carbon dioxide.

9.9 Core Practical — Combustion of Alcohols

Investigate the **temperature rise produced in a known mass of water by the combustion of the alcohols ethanol, propanol, butanol and pentanol**.

Comparing the energy released by burning alcohols



Method

1. Measure a known volume of water (e.g. 100 cm^3 ≈ 100 g) into a copper calorimeter can; clamp it a fixed distance above the burner. Record the starting temperature.
2. Weigh the spirit burner (with lid) containing the alcohol.

3. Light the wick and heat the water, stirring with the thermometer, until the temperature has risen by a set amount (e.g. 20 °C).
4. Extinguish the flame with the lid and immediately reweigh the burner. Mass of alcohol burned = initial mass – final mass.
5. Repeat for each alcohol, keeping the mass of water, distance from flame, wick length and temperature rise the same.
6. Calculate **temperature rise per gram** of alcohol burned (or mass needed for the fixed temperature rise) to compare.

Alcohol	Mass burned / g	Temperature rise / °C	Temperature rise per gram / °C g^{-1}
ethanol	1.10	20.0	18.2
propanol	0.95	20.0	21.1
butanol	0.88	20.0	22.7
pentanol	0.84	20.0	23.8

Illustrative data only. Trend: as the chain gets longer, more energy is released per gram (and much more per mole), because each molecule has more C–H and C–C bonds to break and more CO_2 and H_2O bonds formed.

Worked Example: *(Illustrative data.)* Burning 0.92 g of propanol raises the temperature of 100 g of water from 19.5 °C to 37.0 °C. Find the temperature rise per gram of propanol. Temperature rise = 37.0 – 19.5 = 17.5 °C. Rise per gram = 17.5 ÷ 0.92 = **19 °C g^{-1}** (2 s.f.).

Exam Tip: The measured values are always **lower** than the true energy released because of **heat loss** to the surroundings (and to heating the can), and **incomplete combustion** (soot on the can). Improvements: draught shields, lid on can, keep flame distance fixed and small, stir the water, use a fresh wick. Safety: alcohols are highly flammable — keep stoppered bottles away from flames.

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9.10 Addition Polymers

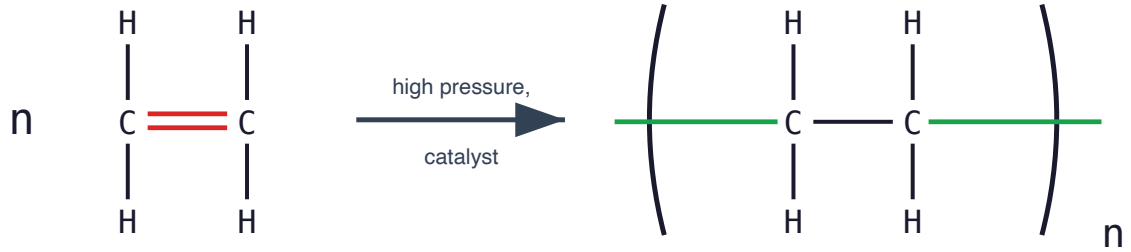
A **polymer** is a substance of **high average relative molecular mass** made up of **small repeating units**. The small molecules that join together are called **monomers**.

Addition polymerisation

Alkenes (and other monomers with a C=C double bond) join together in **addition polymerisation**. The double bond in each monomer **opens up** and the monomers join in a long chain. The polymer is the **only product**, and it is saturated (the C=C bonds are gone).

$$n\ CH_2{=}CH_2 \rightarrow -(CH_2{-}CH_2)_n-$$ ethene → poly(ethene)

Ethene → poly(ethene): drawing the repeat unit



monomer: ethene (C=C)

repeat unit: C=C now single; bonds extend out through the brackets

Other addition polymers

Monomer	Monomer structure	Polymer	Repeat unit
ethene	$CH_2{=}CH_2$	poly(ethene)	$-(CH_2{-}CH_2)-$
propene	$CH_2{=}CHCH_3$	poly(propene)	$-(CH_2{-}CH(CH_3))-$
chloroethene	$CH_2{=}CHCl$	poly(chloroethene), **PVC**	$-(CH_2{-}CHCl)-$
tetrafluoroethene	$CF_2{=}CF_2$	poly(tetrafluoroethene), **PTFE**	$-(CF_2{-}CF_2)-$

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Deducing structures

- **Monomer → polymer:** draw the two C atoms of the C=C as a single bond; keep all the other groups attached in the same positions; add bonds out to each side through square brackets; put *n* after the bracket.
- **Polymer → monomer:** take the two-carbon repeat unit, remove the extension bonds and brackets, and put back a **C=C double bond** between the two carbons.

Worked Example: The repeat unit of a polymer is $-(CH_2-CH(C_6H_5))-$. Draw the monomer. Replace the C–C single bond in the chain by a double bond and remove the extension bonds: **$CH_2=CH(C_6H_5)$** — this is phenylethene (styrene); the polymer is poly(phenylethene), "polystyrene".

Exam Tip: The main chain of an addition polymer is **always carbon atoms only**. Side groups (CH_3, Cl, F) hang off it. A very common error is putting the CH_3 of propene into the main chain — the repeat unit of poly(propene) has only **two** carbons in the backbone.

Uses and properties

Polymer	Properties	Uses
Poly(ethene)	flexible, cheap, tough, easily moulded, electrical insulator	plastic bags, bottles, cling film, cable insulation
Poly(propene)	tough, strong, flexible, high softening point, resistant to chemicals	crates, ropes, carpets, food containers, car bumpers
PVC (poly(chloroethene))	tough, hard-wearing, waterproof; can be rigid or made flexible with plasticisers; electrical insulator	window frames, drainpipes, water pipes, insulation on electrical wires

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PTFE (poly(tetrafluoroethene))	very unreactive, tough, **non-stick** (very low friction), high melting point	non-stick coatings for frying pans, waterproof breathable clothing, lab equipment

Problems with polymers

- **Availability of starting materials:** most monomers come from **crude oil**, a finite resource; as it runs out, polymers will become more expensive.
- **Persistence in landfill:** most addition polymers are **non-biodegradable** — microorganisms cannot break them down — so they persist in landfill sites for hundreds of years and pollute oceans (harming wildlife; microplastics).
- **Gases produced during disposal by combustion:** burning plastics releases energy but produces **CO_2** (greenhouse gas) and can give **toxic gases** — e.g. PVC releases **hydrogen chloride**; incomplete combustion gives CO and soot.

Recycling polymers

Advantages of recycling	Disadvantages / problems
Reduces the use of crude oil (finite resource)	Plastics must be **collected and transported** — costs money and uses fuel
Reduces the amount of waste in landfill	Different polymers must be **sorted** — difficult and labour-intensive
Less CO_2 / toxic gas than burning	Recycled plastic is often lower quality; can only be recycled a limited number of times
Usually uses less energy than making new polymer	Washing and processing uses energy and water

9.11 Condensation Polymers (HT)

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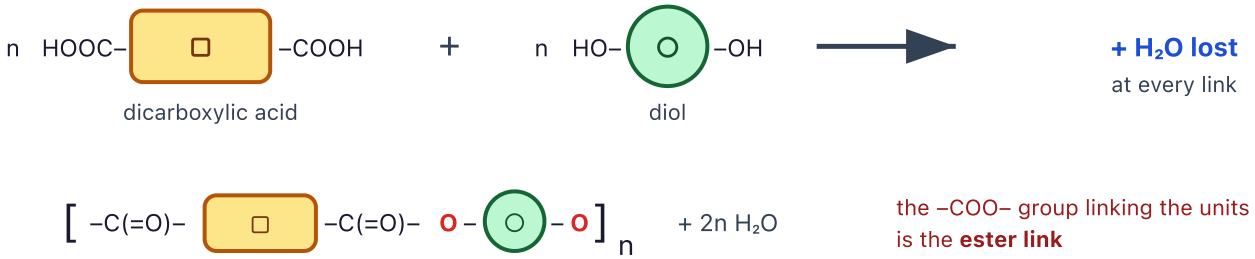
In **condensation polymerisation**, monomers with **two functional groups** join together and a **small molecule (usually water)** is **lost** each time a link forms.

Polyesters

A **polyester** is formed when a monomer with **two carboxylic acid groups** (a dicarboxylic acid) reacts with a monomer with **two alcohol groups** (a diol). Each reaction between –COOH and –OH forms an **ester link** and releases **one molecule of water**.

$$n\ HOOC–□–COOH + n\ HO–○–OH \rightarrow –(OC–□–COO–○–O)_n– + 2n\ H_2O$$

Forming a polyester (□ and ○ represent the carbon chains)



	Addition polymerisation	Condensation polymerisation
Monomers	one type, each with a C=C double bond	usually two different monomers, each with **two** functional groups
Products	polymer only	polymer **and** a small molecule (e.g. water)
Example	poly(ethene), PVC, PTFE	polyesters (e.g. PET for bottles and fibres)

Biodegradable polymers

Some polyesters (e.g. poly(lactic acid), PLA, made from plant starch) are **biodegradable** — they can be broken down by microorganisms, so they cause fewer landfill problems.

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9.12 Naturally Occurring Polymers

Many important biological molecules are polymers:

Natural polymer	Monomers	Notes
DNA	**nucleotides** (four different ones, containing the bases A, T, C, G)	two polymer strands twisted into a double helix; carries genetic information
Proteins	**amino acids** (about 20 different ones)	formed by condensation polymerisation; enzymes, muscle, hair
Carbohydrates — starch, cellulose, glycogen	**sugars** (e.g. glucose)	formed by condensation polymerisation; starch is an energy store in plants, cellulose forms plant cell walls

9.13 Bulk Properties of Materials

Materials are chosen for a job by matching their **properties** to the use. You need to compare the properties of **ceramics, polymers, composites and metals**.

Material	Examples	Typical properties	Uses
Ceramics	glass, clay ceramics (pottery, bricks, porcelain)	hard, strong in compression, **brittle**, high melting point, electrical and thermal insulators, unreactive, often transparent (glass)	windows, bottles, plates, bricks, tiles, insulators on power lines
Polymers	poly(ethene), PVC, nylon	low density, flexible, easily moulded, electrical/thermal insulators, unreactive, often low melting/softening point	packaging, pipes, clothing, insulation

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Composites	fibreglass (glass fibres in a polymer), carbon-fibre reinforced polymer, reinforced concrete, wood	made of a **matrix** (binder) surrounding **reinforcement** fibres/fragments; properties combine the best of both — e.g. strong, stiff and low density	boat hulls, bike frames, aircraft parts, tennis rackets, buildings and bridges
Metals	steel, aluminium, copper	strong, **malleable and ductile**, good conductors of heat and electricity, high density (mostly), shiny; some corrode	wires, bridges, car bodies, saucepans

Worked Example: *(Illustrative data.)* A designer wants a material for a racing-bike frame: it must be strong, stiff and light. Which material is best?

Material	Density / g cm^{-3}	Strength / MPa	Relative cost
steel	7.8	500	low
aluminium alloy	2.7	300	medium
carbon-fibre composite	1.6	600	high
PVC	1.4	50	very low

Carbon-fibre composite has the **highest strength and a low density**, so it gives the strongest, lightest frame — the best choice for racing despite the higher cost. PVC is light but far too weak; steel is strong but heavy; aluminium is a cheaper compromise.

Exam Tip: For "choose a material" questions, **use the data**: quote values, compare them, and link each property to the use (e.g. "low density so the bike is light"). Mention cost or other drawbacks to show evaluation.

9.14 Nanoparticles and Surface Properties

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Nanoscience refers to structures that are **1–100 nm** in size, of the order of a few hundred atoms. **Nanoparticles** are particles in this size range (1 nm = 1×10^{-9} m).

For comparison, atoms are about 0.1–0.5 nm across, so a nanoparticle contains a few hundred atoms; fine dust particles ($PM_{2.5}$) are 100–2500 nm.

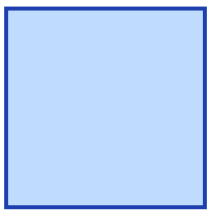
Why nanoparticles behave differently

Nanoparticles show different properties from the same material in bulk because of their **very high surface area to volume ratio**. A much larger fraction of their atoms are on the surface, so:

- they can be much more **reactive** and make very effective **catalysts** (less material needed for the same surface area);
- their optical, electrical and other properties can be different (e.g. nanoparticles of gold appear red or purple, not gold-coloured; nanoparticulate titanium dioxide is transparent rather than white).

As the side length of a cube **decreases by a factor of 10**, the surface area to volume ratio **increases by a factor of 10**.

Smaller particles → larger surface area : volume



side 100 nm
SA : V = 6/100 = 0.06 nm^{-1}



side 10 nm
SA : V = 6/10 = 0.6 nm^{-1}



side 1 nm
SA : V = 6/1 = 6 nm^{-1}

(Cubes not to scale with each other.) For a cube of side L: SA = $6L^2$, V = L^3, so SA : V = 6/L.

Worked Example: Calculate the surface area to volume ratio of a cube-shaped nanoparticle of side 20 nm.
SA = $6 \times 20^2 = 2400$ nm^2; V = $20^3 = 8000$ nm^3; SA : V = 2400 ÷ 8000 = **0.3 nm^{-1}** (or 3 : 10).
A particle of side 2 nm would have SA : V = 6 ÷ 2 = 3 nm^{-1} — ten times larger.

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Applications of nanoparticles

- **Sun creams** — nanoparticles of titanium dioxide or zinc oxide absorb/block UV, give better skin coverage and are transparent (no white residue).
- **Medicine** — targeted drug delivery (nanoparticles can carry drugs into cells); **silver nanoparticles** are antibacterial, used in wound dressings and to coat surgical instruments.
- **Catalysts** — high surface area makes them very efficient (e.g. in catalytic converters).
- **Electronics** — tiny circuits and components; carbon nanotubes conduct electricity.
- **Cosmetics** and self-cleaning glass.

Risks of nanoparticles

- Their properties are **not fully understood**, and long-term effects have not yet been studied — they are a relatively new technology.
- Because they are so small, they may be **breathed in** or **absorbed through the skin** and enter cells and the bloodstream, possibly causing damage.
- Nanoparticles that are washed down drains may harm **aquatic organisms** and the environment (e.g. silver nanoparticles may kill helpful bacteria in sewage works).
- Some people argue products containing nanoparticles should be **tested more thoroughly** and clearly **labelled**.

> **Exam Tip:** For a "discuss the use of nanoparticles" question, give a **benefit linked to high surface area to volume ratio**, then a **risk** (not fully understood, may enter the body/cells, environmental release), and come to a balanced conclusion. Be careful with units: 1 nm = 10^{-9} m, not 10^{-6} m.

Key Points Summary

- **Flame tests:** Li^+ red, Na^+ yellow, K^+ lilac, Ca^{2+} orange-red, Cu^{2+} blue-green. Clean nichrome wire with conc. HCl; mixtures can mask colours.
- **NaOH(aq):** Al^{3+} white ppt (dissolves in excess), Ca^{2+} white ppt (stays), Cu^{2+} blue, Fe^{2+} green, Fe^{3+} orange-brown. NH_4^+: warm with NaOH → NH_3 turns damp red litmus blue.

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- **Anions:** halides — dilute HNO_3 then $AgNO_3$: Cl^- white, Br^- cream, I^- yellow. Sulfate — dilute HCl then $BaCl_2$: white ppt. Carbonate — dilute acid: CO_2 turns limewater milky.
- **Core practical:** identify cation and anion in unknown salts by combining these tests; **(HT)** write ionic equations for precipitates.
- Instrumental methods improve sensitivity, accuracy and speed. A flame photometer identifies ions by comparing with reference spectra (works for mixtures) and finds concentrations from a calibration curve.
- Alkanes (C_nH_{2n+2}: methane, ethane, propane, butane) are saturated; alkenes (C_nH_{2n}: ethene, propene, butene) are unsaturated with a C=C functional group.
- Alkenes decolourise bromine water (orange → colourless) by addition; complete combustion oxidises hydrocarbons to CO_2 + H_2O.
- Alcohols have –OH (methanol, ethanol, propan-1-ol, butan-1-ol). Ethanol is made by fermenting sugar with yeast (30–40 °C, no air), then concentrated by fractional distillation.
- Ethanol can be oxidised (by microbes or chemicals) to ethanoic acid. Carboxylic acids have –COOH (methanoic, ethanoic, propanoic, butanoic); they are weak acids and react with carbonates to give salt + water + CO_2.
- **Core practical:** burn ethanol, propanol, butanol, pentanol to heat a known mass of water; compare temperature rise per gram; longer chain → more energy released. Heat loss and incomplete combustion make values low.
- A polymer has high average relative molecular mass made of small repeating units. Addition polymers: poly(ethene), poly(propene), PVC, PTFE — know how to go from monomer to repeat unit and back, and link uses to properties.
- Polymer problems: crude-oil starting materials, non-biodegradable in landfill, toxic gases when burned; recycling helps but needs collection and sorting.
- **(HT)** Polyesters are condensation polymers of a dicarboxylic acid + diol, with water lost at each ester link.
- DNA (nucleotides), proteins (amino acids), starch and cellulose (sugars) are natural polymers.
- Compare ceramics (hard, brittle, insulating), polymers (light, flexible, insulating), composites (matrix + reinforcement, e.g. strong and light) and metals (strong, malleable, conductive) to choose materials for a use.
- Nanoparticles are 1–100 nm; very high surface area to volume ratio gives different properties. Uses: sun creams, catalysts, medicine, antibacterial silver; risks: not fully understood, may

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enter cells, environmental effects.

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